Examiner only

7. (a) **Describe** a controlled nuclear fission reaction that takes place inside a nuclear reactor and **compare** it to nuclear fusion reactions that occur in stars such as the Sun. You should include a comparison of the two types of reactions as a source of generating electricity. [6 QER]

(b) Complete the equation for the fusion reaction shown below. [2]

$$^{2}_{1}\mathrm{H} \; + \; \ldots\ldots\ldots\ldots \mathrm{H} \longrightarrow \; ^{4}_{2}\mathrm{He} \; + \; ^{1}_{0}\mathrm{n}$$

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